

1. Administrative & Study Identification

Full Study Title: A Phase 2A, Double-blind, Randomized, Placebo-controlled, Multi-center Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of BMS-986435/MYK-224 in Participants with Heart Failure with Preserved Ejection Fraction (HFpEF) **Short Title/Acronym:** AURORA-HFpEF **Protocol Number:** CV029-1001 **NCT Number:** NCT06122779 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** BMS-986435 / MYK-224 (Myosin Modulator) **Phase of Development:** Phase II **Medical Monitor:** Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and exposure-response (E-R) of BMS-986435/MYK-224 in participants with symptomatic Heart Failure with Preserved Ejection Fraction (HFpEF). **Secondary Objectives:** To evaluate the pharmacokinetics (PK) and pharmacodynamics (PD) of BMS-986435/MYK-224 and to assess key efficacy metrics and overall patient outcomes.

2.2 Background & Rationale To address the unmet medical need in patients with stable, symptomatic Heart Failure with Preserved Ejection Fraction (HFpEF). Many patients continue to experience debilitating symptoms such as fatigue and shortness of breath despite taking standard medications. BMS-986435/MYK-224 is a novel myosin modulator designed to help the heart muscle relax better and use less energy, which may improve cardiac function and alleviate HFpEF symptoms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 208 participants **Number of Sites:** Approximately 119 locations globally across multiple countries (e.g., USA, Canada, Spain, Italy) **Estimated Study Start:** 07-Nov-2023 **Estimated Primary Completion:** 03-Jul-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 40 to 85 years. 2. Adult participants with stable, symptomatic Heart Failure with Preserved Ejection Fraction (HFpEF) with a normal heart pumping ability (LVEF $\geq 55\%$). 3. Presence of abnormal heart structural markers, such as thick heart walls or an enlarged upper chamber, and elevated stress chemicals.

4.2 Exclusion Criteria 1. Known diagnosis of obstructive or genetic hypertrophic cardiomyopathy. 2. Infiltrative or storage disorders such as cardiac amyloidosis. 3. Any other acute or serious condition that could interfere with assessments during the study or pose a risk to the participant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administration of BMS-986435/MYK-224 tablets or matching placebo. **Route of Administration:** Oral **Duration of Treatment:** Up to approximately 24 weeks of exposure.

5.2 Concomitant & Prohibited Therapy Permitted: Stable standard-of-care heart failure medications as prescribed by standard medical guidelines. **Prohibited:** Concurrent administration of other investigational therapies or treatments specifically contraindicated in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Echocardiogram, Laboratory Tests
Baseline	Day 0	Randomization, First Dose, Vital Signs
Interim	Weeks 4 to 20	Safety Labs, PK/PD Assessment, Efficacy and Tolerability Evaluation
End of Study	Week 24	Final Vital Signs, Exit Interview, IP Accountability, Final Safety Evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs) up to approximately 24 weeks. **Measurement Method:** Clinical observation, diagnostic tests, and laboratory analysis.

7.2 Secondary & Exploratory Endpoints Exposure-response (E-R) relationships. Pharmacokinetics and Pharmacodynamics (PK/PD) assessments.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of TEAEs via clinical assessments, patient reporting, and onsite visits.

Serious Adverse Events (SAEs): Reported within 24 hours of site awareness, tracked continuously up to approximately 24 weeks.

Data Monitoring Committee (DMC): Independent Data Monitoring Committee appointed by the sponsor to oversee patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for exposure/efficacy metrics; Safety Set for all TEAE/SAE evaluations. **Sample Size**

Justification: A sample size of $N=208$ is powered to provide robust safety, tolerability, and initial dose-finding insights for Phase 2. **Handling of Missing Data:** Appropriate imputation methods such as Multiple Imputation or Mixed-Effects Model for Repeated Measures (MMRM) applied for continuous efficacy endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote clinical monitoring to ensure patient safety and protocol adherence. **Audit Trail:**

Robust tracking via Electronic Data Capture (EDC) systems, eCRF locking, and resolution of data clarification forms.

11. Study Identifiers & Governance

Primary ID: CV029-1001 **Registry Number:** NCT06122779 **Sponsor/Lead**

Organization: Bristol-Myers Squibb **Study Title:** AURORA-HFpEF

Official Regulatory Title: A Phase 2A, Double-blind, Randomized, Placebo-controlled, Multi-center Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of BMS-986435/MYK-224 in Participants with Heart Failure with Preserved Ejection Fraction (HFpEF)

12. Clinical Framework (Heart Failure Specific)

Primary Condition: Heart Failure with Preserved Ejection Fraction (HFpEF) **Diagnosis Threshold:** Normal heart pumping ability ($\geq 55\%$) with symptomatic presentation (NYHA Class II or III)

Medical Methodology: Myosin Modulator Therapy **Optimization**

Target: Improvement of cardiac muscle relaxation, energy utilization, and reduction of HF-related symptoms

13. Study Procedures & Methodology

Management Pathway: Standard HFpEF clinical pathway supplemented with BMS-986435/MYK-224 or placebo. **Education Protocol (HCP):** Clinical site initiation, echocardiogram protocol training, and AE reporting guidelines. **Education Protocol (Patient):** Informed consent process, drug administration compliance, and symptom reporting. **Quality Audit Mechanism:** Routine sponsor oversight, centralized echocardiogram reviews, and site-level adherence monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Up to 24 weeks **Onsite Visit Cadence:** Periodic visits designated throughout the 24-week treatment period for vital assessments. **Remote Monitoring Frequency:** Intermittent telehealth and safety follow-ups as determined by local protocols. **Checklist Review Interval:** Regular reconciliation of concomitant medications and adverse events.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				